

Unit Rates with Whole Numbers

Answer Key

Grade 6

Quick Answers

- | | |
|------------------------------|---|
| 1. 3 | 8. (a) dollars per pen, offer A = 4, (b) dollars per pen, offer B = 3, (c) better deal = > |
| 2. 20 | 9. (a) car A, miles per gallon = 30, (b) car B, miles per gallon = 35, (c) farther per gallon = < |
| 3. 56 | 10. (a) 6-pack, dollars per can = 2, (b) 10-pack, dollars per can = 2, (c) comparison = =, — |
| 4. 3 | |
| 5. 50 mph | |
| 6. 1 hour = 12, 5 hours = 60 | |
| 7. 7 oz | |
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What is verified

9 of 10 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problem 10. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 6

6.RP.A.2 – problems 1, 2, 4, 5, 7, 8, 9, 10

6.RP.A.3 – problems 3, 6

Difficulty 1–4 of 5 across 18 tagged checks.

Common wrong answers

5. *If they got 0.02: divided hours by miles instead of miles by hours*

9. *If they got 0.03: divided gallons by miles instead of miles by gallons*

1. 12 eggs make 4 equal pancake batches. Eggs per batch?

Solution: Sharing 12 eggs equally among 4 groups is the division $12 \div 4 = 3$ — each circle in the model ends up with the same count. The unit rate is 3 eggs per batch. 3

2. 60 pages in 3 hours. Pages in one hour?

Solution: To go from 3 hours to one hour, divide both rows of the table by 3: $60 \div 3 = 20$ pages per hour. 20

3. One ticket costs 8 dollars. Cost of 7 tickets?

Solution: The unit rate is already given (8 dollars per ticket), so multiply by the number of tickets: $8 \times 7 = 56$ dollars.

56

4. 6 comic books cost 18 dollars. Dollars per book?

Solution: Unit price = total cost $\div$ number of books: $18 \div 6 = 3$ dollars per book.

3

5. 150 miles in 3 hours. Speed as a unit rate?

Solution: “Miles per hour” puts miles on top: $150 \div 3 = 50$.

50 mph

6. 24 posters every 2 hours. Complete the table for 1 hour and 5 hours.

Solution: Unit rate first: $24 \div 2 = 12$ posters in 1 hour.

12

Then use the rate: $12 \times 5 = 60$ posters in 5 hours.

60

7. A 28-oz jar fills 4 equal serving bowls. Ounces per serving?

Solution: $28 \div 4 = 7$ ounces in one serving.

7 oz

8. Offer A: 3 pens for 12 dollars. Offer B: 5 pens for 15 dollars. Which is the better deal?

Solution: (a) Offer A: $12 \div 3 = 4$ dollars per pen.

4

(b) Offer B: $15 \div 5 = 3$ dollars per pen.

3

(c) $4 > 3$, so offer B costs less per pen — B is the better deal.

$4 > 3$, offer B

9. Car A: 240 miles on 8 gallons. Car B: 210 miles on 6 gallons. Which goes farther per gallon?

Solution: (a) Car A: $240 \div 8 = 30$ miles per gallon.

30

(b) Car B: $210 \div 6 = 35$ miles per gallon.

35

(c) $30 < 35$, so car B goes farther on one gallon.

$30 < 35$, car B

10. 6-pack for 12 dollars vs 10-pack for 20 dollars. Compare unit prices and interpret.

Solution: (a) 6-pack: $12 \div 6 = 2$ dollars per can.

2

(b) 10-pack: $20 \div 10 = 2$ dollars per can.

2

(c) The unit prices are equal.

$2 = 2$

(d) **Interpretation (instructor judges):** Full credit says both packs charge 2 dollars per can, so neither is a better deal per can — a shopper can choose by how many cans they need, not by price per can.